

QENA SOUTH VALLEY UNIVERSITY, Egypt

WMO: 624030

Lat: 26.2003N

Lon: 32.7467E

Elev: 77

StdP: 100.40

Time Zone: 2.00 (E02)

Period: 04-17

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	7.1	8.2	-10.8	1.5	27.3	-8.6	1.8	26.9	7.6	16.1	6.7	17.3	1.1	0	0.402

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.8	42.7	19.0	41.6	19.0	40.6	19.0	22.5	37.1	21.6	36.5	20.9	36.0	4.0	350

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.1	12.3	30.8	15.7	11.3	30.2	14.7	10.5	29.9	66.4	37.0	63.0	36.2	60.5	36.1	28.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
7.8	6.9	6.2	5.1	44.2	1.3	1.3	4.1	45.2	3.4	46.0	2.6	46.7	1.7	47.7		
			1.2	25.0	1.2	1.4	0.4	26.0	-0.3	26.8	-1.0	27.6	-1.8	28.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	25.9	15.2	18.1	21.7	26.9	30.4	33.2	33.5	33.2	31.2	27.9	21.7	16.8
	DBStd	7.05	2.67	3.26	3.26	3.32	2.85	2.00	1.50	1.43	1.91	2.85	3.09	2.66
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	219	106	40	5	0	0	0	0	0	0	0	4	64
	CDD10.0	5787	161	227	364	506	632	696	728	721	636	556	351	210
	CDD18.3	2964	8	33	111	256	373	446	469	462	386	298	105	16
	CDH23.3	43341	83	358	1188	3299	5641	7370	7736	7455	5610	3574	893	134
	CDH26.7	27089	8	112	474	1842	3577	5047	5284	5016	3464	1931	306	28
Wind	WSAvg	2.6	1.9	2.2	2.5	2.8	2.9	3.2	3.2	3.2	3.1	2.3	2.0	1.9
Precipitation	PrecAvg	1	0	0	0	0	1	0	0	0	0	0	0	0
	PrecMax	12	2	2	3	7	12	0	0	0	0	0	0	1
	PrecMin	0	0	0	0	0	0	0	0	0	0	0	0	0
	PrecStd	3	0	0	1	1	2	0	0	0	0	0	0	0
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	27.8	32.8	37.1	40.4	43.5	44.8	43.2	42.7	41.6	40.1	34.4	29.8
		MCWB	15.8	17.2	17.4	17.8	18.6	18.9	19.6	19.9	19.1	18.7	17.2	15.8
	2%	DB	25.9	30.0	33.9	38.5	41.2	43.1	41.8	41.3	40.2	37.9	32.1	26.8
		MCWB	14.5	15.6	15.8	17.0	18.1	18.7	19.5	19.8	19.3	18.7	17.5	15.4
	5%	DB	23.5	27.5	31.2	36.9	39.7	41.7	40.7	40.4	38.8	36.2	30.1	24.5
		MCWB	12.9	14.3	14.6	16.4	17.6	18.5	19.5	19.6	19.1	18.7	16.9	14.0
	10%	DB	21.4	25.2	29.1	35.0	38.1	40.2	39.7	39.5	37.6	34.7	28.1	22.7
		MCWB	12.0	13.1	14.3	15.8	16.9	18.2	19.3	19.6	19.2	18.5	16.0	13.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	18.7	17.9	20.0	20.1	19.7	21.3	23.8	24.0	23.2	21.9	20.1	17.8
		MCDB	25.1	31.2	28.7	33.2	37.9	38.7	38.6	38.5	35.6	34.4	30.3	26.0
	2%	WB	15.2	16.2	17.4	18.2	18.9	20.2	22.2	22.5	21.7	20.6	18.4	15.9
		MCDB	23.7	27.5	30.9	35.0	38.9	37.9	37.7	37.4	35.7	33.8	29.2	24.2
	5%	WB	13.6	14.7	15.8	17.2	18.2	19.5	21.2	21.5	20.9	19.8	17.5	14.7
		MCDB	21.8	26.3	30.2	34.9	37.7	37.8	36.8	36.8	35.2	32.8	28.0	23.2
	10%	WB	12.5	13.5	14.5	16.3	17.5	18.9	20.5	20.7	20.2	19.1	16.5	13.7
		MCDB	20.4	24.2	28.0	33.7	36.4	37.6	36.4	36.3	34.4	32.5	27.0	21.8
Mean Daily Temperature Range	5% DB	MDBR	12.2	12.9	13.4	14.0	13.9	13.8	12.8	13.0	13.1	12.7	12.4	12.2
		MCDBR	14.0	15.1	15.4	15.3	15.6	15.2	13.6	13.7	14.1	13.7	13.7	13.6
		MCWBR	7.7	7.2	6.7	6.0	5.4	4.8	4.0	4.3	4.4	4.9	6.1	6.9
	5% WB	MCDBR	12.9	14.3	13.9	14.2	14.4	13.9	12.6	12.9	13.0	12.7	12.8	12.9
	MCWBR	7.6	7.1	6.7	6.1	5.2	4.7	4.6	5.1	4.7	4.8	5.9	6.7	
Clear-Sky Solar Irradiance	taub	0.368	0.400	0.443	0.527	0.541	0.461	0.469	0.470	0.456	0.458	0.384	0.365	
	taud	2.194	2.077	1.948	1.762	1.748	1.967	1.977	1.980	1.997	1.973	2.171	2.212	
	Ebn at Noon	870	868	850	789	775	834	827	826	827	798	844	856	
	Edh at Noon	127	154	185	229	232	185	183	181	174	168	129	120	
All-Sky Solar Radiation	RadAvg	4.44	5.38	6.51	7.34	7.79	8.17	7.91	7.45	6.72	5.63	4.71	4.16	
	RadStd	0.16	0.18	0.20	0.14	0.20	0.08	0.10	0.11	0.15	0.11	0.09	0.07	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (4 neighbors)	+0.57	N/A	N/A	+0.57	+0.68	+0.64	N/A	-35	+232	+205

Nomenclature: See separate page